ISDB-BISEW IT SCHOLARSHIP PROGRAMME

COURSE PROJECT PROPOSAL

Pharmacy Sales & Inventory Management System

SUBMITTED BY
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1. PROJECT OVERVIEW & BACKGROUND

In today's fast-paced healthcare environment, traditional manual record-keeping in retail pharmacies is highly prone to human error, creates stock mismatches, risks selling expired medicines, and causes substantial delays in customer billing. The **Pharmacy Sales & Inventory Management System** is a centralized web application designed to automate pharmacy operations, streamline stock control, digitize Point of Sale (POS) invoicing, and secure transactional records.

2. CORE OBJECTIVES

- **Digitize Sales Operations:** Replace manual paper invoices with an automated Point-of-Sale (POS) cashier interface to perform rapid checkout and billing.
- **Real-Time Inventory Control:** Maintain precision stock tracking for items received, sold, and available, integrated with automated low-stock and expiry date alerts.
- **Role-Based Access & Security:** Centralize records within a relational MySQL database with strict role-based access control (Admin, Cashier, Staff).
- **Business Decision Reporting:** Generate automated summaries for daily revenue, stock turnover, and supplier audit trails to guide pharmaceutical operations.

3. TECHNOLOGY STACK & TOOLS

Component	Technology / Tool	Purpose in Project
Backend Logic	Core PHP (8.2+ / 8.3 Latest)	Server-side request handling, session control, business logic & security.
Database Engine	MySQL / MariaDB (via PDO)	Relational schema, indexed tables, multi-table queries & data integrity.
Frontend UI	HTML5, CSS3, JavaScript, Bootstrap	Responsive user interface, interactive dynamic forms & modal dialogs.
Local Server Tool	XAMPP Environment	Apache local web server & MariaDB database engine hosting.
Development Tool	Notepad++ / VS Code	Source code authoring, script formatting, and file structure management.

4. ROLE-BASED ACCESS CONTROL (RBAC)

User Role	Access Privileges	Key Responsibilities
Administrator	Full Access (All Modules)	Manage user accounts, set up categories/types, approve receiving, override inventory, and review financial reports.
Cashier	Sales POS & Customer Records	Process sales transactions, calculate bill totals, accept payments, issue customer receipts, and track change amounts.
Store Staff	Inventory & Receiving Access	Log stock receiving from suppliers, track batch expiry dates, audit low-stock items, and manage product lists.

5. COMPREHENSIVE MODULE BREAKDOWN

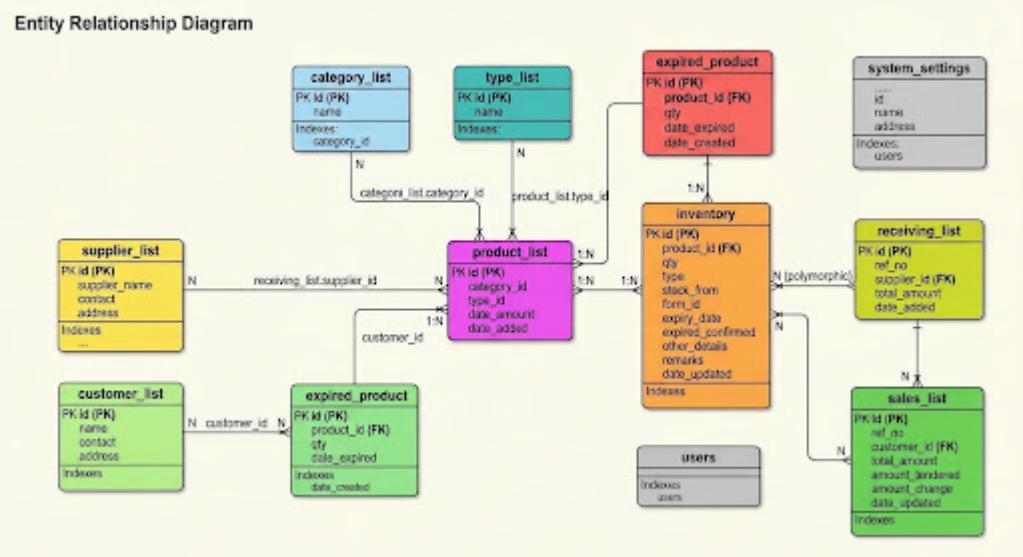
Module Name	Key Parameters / Fields	Functional Features
Sales (POS)	Ref No, Customer, Product, Quantity, Unit Price, Total, Amount Tendered, Change	Dynamic order list, real-time total computation, cash receipt generation, automated stock deduction upon payment.
Inventory Log	Product Name, Stock In, Stock Out, Expired Qty, Stock Available, Expiry Date	Central ledger for movements; search filter, entry pagination, real-time current stock calculation.
Receiving Stock	Date, Reference No, Supplier Name, Items Received, Total Amount, Action	Record batch inventory purchases from suppliers; automatically updates main inventory stock numbers.
Medicine Master	SKU, Product Name, Category, Type/Form, Measurement (e.g. 500mg), Price, Prescription Req.	Master catalogue management; add, edit, or archive pharmaceutical items with unit attributes.
Category & Type	Category Name, Formulation Type (Capsule, Tablet, Liquid, Inhaler, Drops)	Classification system for fast filtering and organized pharmaceutical indexing.
Expired Products	Product ID, Quantity, Date Expired, Confirmation Status, Date Logged	Isolate past-expiry medicine batches to prevent accidental prescription or sale.
Supplier & Customer	Name, Contact Number, Address Details, Order Reference History	Directory for commercial vendor purchases and customer relationship management.
User Management	Full Name, Username, Password (Hashed), User Role (Admin, Cashier, Staff)	Secure credential management with session-based permission control.

6. RELATIONAL DATABASE SCHEMA OVERVIEW

The system relies on 10 relational tables designed for operational efficiency. Inventory logs utilize a dynamic reference strategy connecting stock events directly with `receiving_list` or `sales_list` tables.

- `product_list` $\rightarrow$ Linked to `category_list` and `type_list` via `category_id` and `type_id`.
- `inventory` $\rightarrow$ Central ledger containing `product_id`, `stock_from` ('receiving' or 'Sales'), and `form_id`.
- `receiving_list` $\rightarrow$ Relates stock receipts to `supplier_list` through `supplier_id`.
- `sales_list` $\rightarrow$ Connects customer billing records to `customer_list` through `customer_id`.
- `expired_product` $\rightarrow$ Stores specific stock units confirmed expired from batch logs.

7. ENTITY-RELATIONSHIP (ER) DIAGRAM



8. CORE APPLICATION WORKFLOWS

A. Stock Receiving & Inventory Update Flow

1. Staff initiates a **New Receiving** entry, selecting the supplier and entering purchase reference numbers.
2. Items are added with batch quantities, cost prices, and batch expiry dates.
3. Upon saving, the backend inserts a row in `receiving_list` and logs positive entry types in the `inventory` table.
4. Available stock metrics update immediately across the entire system catalog.

B. Sales Checkout (POS) Flow

1. Cashier selects customer (or defaults to walk-in) and scans/selects products.
2. System verifies available stock in real time before allowing item addition.
3. Cashier enters customer tendered amount; system calculates exact change.
4. Transaction completes: records written to `sales_list`, inventory stock automatically debited, and printable receipt rendered.

9. SECURITY & DATA INTEGRITY PROTOCOL

Security Aspect	Implementation Standard
SQL Injection Prevention	All database interactions utilize PDO Prepared Statements with strict parameter binding.
Credential Security	User passwords are stored securely using standard `password_hash()` encryption.
Session & Access Control	Route protection middleware checks user login status and role permissions prior to controller rendering.
Data Validation	Strict server-side validation for numeric quantities, unit prices, and valid date entries.

10. EXPECTED PROJECT DELIVERABLES

Upon completion, the **Pharmacy Sales and Inventory Management System** will serve as a complete, industry-ready application for modernizing retail pharmacy management. It will substantially reduce manual administrative workload, prevent revenue loss from unmonitored medicine expiry, accelerate sales transactions, and guarantee accurate stock accountability.

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PROGRAMME AUTHORITY APPROVAL

Signature & Date